

Notes on Regular Epimorphisms and the Monadicity of the Pullback Functor in an Exact Category

A. Cigoli and C. D'Ascoli

These notes are based on the work of Sobral and Tholen [ST92]. This version includes revised proofs adapted to the specific needs of our case. Some ideas are drawn from Janelidze and Tholen [JT91]. Our goal is to prove the following theorem.

Theorem. In an exact category, if $p : E \rightarrow B$ is a regular epi, then p^* is a monadic functor.

We recall that a right adjoint is monadic if the comparison functor $K : (\mathcal{A} \downarrow B) \rightarrow (\mathcal{A} \downarrow E)^T$ is an equivalence of categories, where $T = p^*p_!$ is the monad associated to p .

$$\begin{array}{ccc}
 (\mathcal{A} \downarrow B) & \xrightarrow{K} & (\mathcal{A} \downarrow E)^T \\
 \uparrow p_! \quad \downarrow p^* & \nearrow F^T & \\
 (\mathcal{A} \downarrow E) & \xleftarrow{U^T} &
 \end{array}$$

Let $\mathcal{A}$ be a category with pullbacks and let $p : E \rightarrow B$ be a morphism in $\mathcal{A}$. Let $(C, \gamma; \xi)$ be an object in $(\mathcal{A} \downarrow E)^T$ with $\xi : T(C) \rightarrow C$ an algebra.

Consider the pullback in $\mathcal{A}$.

$$\begin{array}{ccc}
 E \times_B C & \xrightarrow{\text{proj}_{2,C}} & C \\
 \text{proj}_{1,C} \downarrow & \nearrow \gamma & \downarrow p\gamma \\
 E & \xrightarrow{p} & B
 \end{array}$$

Thus, we have the span in $\mathcal{A}$:

$$\begin{array}{ccc}
 & E \times_B C & \\
 \text{proj}_{2,C} \swarrow & & \searrow \xi \\
 C & & C
 \end{array}$$

Since ξ is also a morphism in $(\mathcal{A} \downarrow E)$, we have that $\gamma\xi = \text{proj}_{1,C}$

$$\begin{array}{ccc}
 E \times_B C & \xrightarrow{\xi} & C \\
 \text{proj}_{1,C} \searrow & & \swarrow \gamma \\
 & E &
 \end{array}$$

In the following for the p^* -image of a morphism f in $(\mathcal{A} \downarrow B)$ we will use both notation $p^*(f)$ and $1_E \times_C f$.

We recall that the unit η of the monad in an object (C, γ) in $(\mathcal{A} \downarrow E)$ is $\langle \gamma, 1_C \rangle$ and the counit ϵ in an object (D, ζ) in $(\mathcal{A} \downarrow B)$ is the morphism $\text{proj}_{2,D} : E \times_B D \rightarrow D$ given by the pullback of p along $\zeta : D \rightarrow B$.

Lemma 1. *Let $\mathcal{A}$ be a finitely complete category and let $p : E \rightarrow B$ be a morphism in $\mathcal{A}$. For any object $(C, \gamma; \xi)$ in $(\mathcal{A} \downarrow E)^T$, with T the monad associated to p , the morphism ξ is the coequalizer of $(1_E \times_B \xi, 1_E \times_B \text{proj}_{2,C})$.*

Proof. Let $(C, \gamma; \xi)$ be an object in $(\mathcal{A} \downarrow E)^T$. First we aim to show that $\xi(1_E \times_B \xi) = \xi(1_E \times_B \text{proj}_{2,C})$. This holds because ξ is an algebra for the monad $T = (p^*p_!, p^*\epsilon p_!, \eta)$. Observe that the multiplication in an object (C, γ) is

$$p^*\epsilon p_!(C, \gamma) = p^*(\epsilon_{(C, p\gamma)}) = p^*(\text{proj}_{2,C}) = 1_E \times_B \text{proj}_{2,C}$$

Then we have the algebra property

$$\begin{array}{ccc} E \times_B (E \times_B C) & \xrightarrow{1_E \times_B \xi} & E \times_B C \\ \downarrow 1_E \times_B \text{proj}_{2,C} & & \downarrow \xi \\ E \times_B C & \xrightarrow{\xi} & C \end{array}$$

$$\xi(1_E \times_B \text{proj}_{2,C}) = \xi(1_E \times_B \xi)$$

It remains to prove the universal property of the coequalizer. First we observe that ξ is a split epi, in fact $\xi\eta_{(C,\gamma)} = 1_C$. Thus ξ is the coequalizer of $(1_{E \times_B C}, \eta_{(C,\gamma)}\xi)$. We aim to prove that ξ is also the coequalizer of $(1_E \times_B \xi, 1_E \times_B \text{proj}_{2,C})$. We already know that the universal property holds for ξ with $(1_{E \times_B C}, \eta_{(C,\gamma)}\xi)$, so it suffices to show that there exist a morphism $t : E \times_B C \rightarrow E \times_B (E \times_B C)$ such that

- $(1_E \times_B \xi)t = \eta_{(C,\gamma)}\xi$
- $(1_E \times_B \text{proj}_{2,C})t = 1_{E \times_B C}$

Then given a morphism f such that $f(1_E \times_B \text{proj}_{2,C}) = f(1_E \times_B \xi)$, we have, composing with t ,

$$\begin{aligned} f(1_E \times_B \text{proj}_{2,C})t &= f(1_E \times_B \xi)t \\ \implies f1_{E \times_B C} &= f\eta_{(C,\gamma)}\xi \end{aligned}$$

By the universal property for ξ with $(1_{E \times_B C}, \eta_{(C,\gamma)}\xi)$ we obtain the universal property

for ξ with $(1_E \times_B \xi, 1_E \times_B \text{proj}_{2,C})$.

$$\begin{array}{ccccc}
 & & E \times_B C & & \\
 & \swarrow t & \downarrow 1_{E \times_B C} & \downarrow \eta_{(C,\gamma)} \xi & \\
 E \times_B (E \times_B C) & \xrightarrow[1_E \times_B \text{proj}_{2,C}]{1_E \times_B \xi} & E \times_B C & \xrightarrow{f} & F \\
 & & \downarrow \xi & \nearrow \exists! g & \\
 & & C & &
 \end{array}$$

We now have to prove that such a morphism t actually exists. Define $t = \langle \text{proj}_{1,C}, 1_{E \times_B C} \rangle$. We aim to prove that this morphism satisfies the two conditions above, ie the following diagram of pairs commutes.

$$\begin{array}{ccc}
 & E \times_B C & \\
 \langle \text{proj}_{1,C}, 1_{E \times_B C} \rangle \swarrow & \downarrow 1_{E \times_B C} & \downarrow \eta_{(C,\gamma)} \xi \\
 E \times_B (E \times_B C) & \xrightarrow[1_E \times_B \text{proj}_{2,C}]{1_E \times_B \xi} & E \times_B C
 \end{array}$$

In order to prove that we use the jointly monic pair $(\text{proj}_{1,C}, \text{proj}_{2,C})$. Observe that the following compositions hold.

- $\text{proj}_{2,C}(1_E \times_B \xi) \langle \text{proj}_{1,C}, 1_{E \times_B C} \rangle = \xi \text{proj}_{2,E \times_B C} \langle \text{proj}_{1,C}, 1_{E \times_B C} \rangle = \xi 1_{E \times_B C} = \xi$
- $\text{proj}_{2,C} \eta_{(C,\gamma)} \xi = \text{proj}_{2,C} \langle \gamma, 1_C \rangle \xi = 1_C \xi = \xi$
- $\text{proj}_{2,C}(1_E \times_B \text{proj}_{2,C}) \langle \text{proj}_{1,C}, 1_{E \times_B C} \rangle = \text{proj}_{2,C} \text{proj}_{2,E \times_B C} \langle \text{proj}_{1,C}, 1_{E \times_B C} \rangle = \text{proj}_{2,C} 1_{E \times_B C} = \text{proj}_{2,C}$
- $\text{proj}_{2,C} 1_{E \times_B C} = \text{proj}_{2,C}$
- $\text{proj}_{1,C}(1_E \times_B \xi) \langle \text{proj}_{1,C}, 1_{E \times_B C} \rangle = \text{proj}_{1,E \times_B C} \langle \text{proj}_{1,C}, 1_{E \times_B C} \rangle = \text{proj}_{1,C}$
- $\text{proj}_{1,C} \eta_{(C,\gamma)} \xi = \text{proj}_{1,C} \langle \gamma, 1_C \rangle \xi = \gamma \xi = \text{proj}_{1,C}$
- $\text{proj}_{1,C}(1_E \times_B \text{proj}_{2,C}) \langle \text{proj}_{1,C}, 1_{E \times_B C} \rangle = \text{proj}_{1,E \times_B C} \langle \text{proj}_{1,C}, 1_{E \times_B C} \rangle = \text{proj}_{1,C}$
- $\text{proj}_{1,C} 1_{E \times_B C} = \text{proj}_{1,C}$.

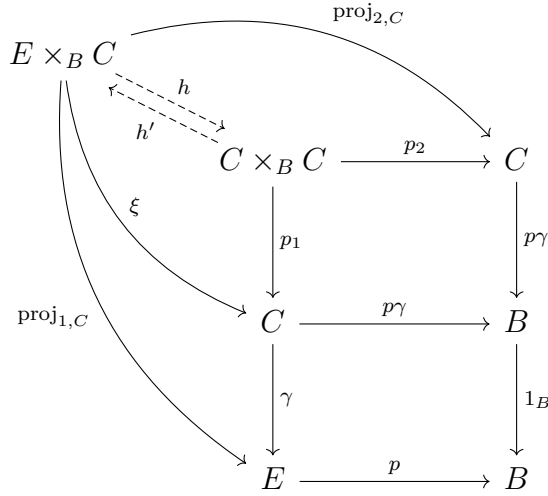
Thus $t = \langle \text{proj}_{1,C}, 1_{E \times_B C} \rangle$ satisfies the conditions required. $\square$

Lemma 2. *Let $\mathcal{A}$ be a finitely complete category and let $p : E \rightarrow B$ be a morphism in $\mathcal{A}$. For any object $(C, \gamma; \xi)$ in $(\mathcal{A} \downarrow E)^T$, with T the monad associated to p , the pair $(\text{proj}_{2,C}, \xi)$ is an equivalence relation in $\mathcal{A}$.*

Proof. First we show that $(\text{proj}_{2,C}, \xi)$ is an internal relation. We need to show that $\langle \text{proj}_{2,C}, \xi \rangle$ is a mono. Consider the following diagram of solid arrows¹. It commutes, in fact

- $(p\gamma)\text{proj}_{2,C} = 1_B(p\gamma)\text{proj}_{2,C} = p\text{proj}_{1,C} = p\gamma\xi$
- $1_B(p\gamma)p_2 = 1_B(p\gamma)p_1 = p\gamma p_1$

Thus we can consider the morphism h' induced by the universal property of the pullback from p_2 and γp_1 , and h similarly induced from $\text{proj}_{2,C}$ and ξ .



In order to prove that $\langle \text{proj}_{2,C}, \xi \rangle$ is a mono, we first need to show that h is a mono. Since $(\text{proj}_{1,C}, \text{proj}_{2,C})$ is a jointly monic pair, by the following compositions

- $\text{proj}_{1,C}h'h = \gamma p_1 h = \gamma \xi = \text{proj}_{1,C} = \text{proj}_{1,C}1_{E \times_B C}$
- $\text{proj}_{2,C}h'h = p_2 h = \text{proj}_{2,C} = \text{proj}_{2,C}1_{E \times_B C}$

we have that $h'h = 1_{E \times_B C}$. This implies that h is a mono, in fact suppose that f and g are two morphisms such that $hf = hg$, then we have

$$\begin{aligned}
 hf &= hg \\
 \implies h'hf &= h'hg \\
 \implies 1_{E \times_B C}f &= 1_{E \times_B C}g \\
 \implies f &= g
 \end{aligned}$$

¹[ST92] assumes the existence of coequalizers, which is not the case in our setting. However, this is not a problem: at this point, we can use the morphism $p\gamma$ instead of the coequalizer of $(\text{proj}_{2,C}, \xi)$, as it serves our purpose. Then, *a posteriori*, this proposition ensures, under the assumption of exact category, that the pair $(\text{proj}_{2,C}, \xi)$ has a coequalizer, which we will make use of.

Consider now the canonical morphism $i : C \times_B C \rightarrow C \times C$ induced by the universal property of the product. We have that i is a mono, in fact if f and g are two morphisms such that $if = ig$, then we have

$$\begin{array}{ccccc}
 & & & C & \\
 & & p_1 \nearrow & \uparrow p'_1 & \\
 A & \xrightarrow[f]{g} & C \times_B C & \xrightarrow{i} & C \times C \\
 & & p_2 \searrow & \downarrow p'_2 & \\
 & & & C &
 \end{array}$$

- $if = ig \implies p'_1 if = p'_1 ig \implies p_1 f = p_1 g$
- $if = ig \implies p'_2 if = p'_2 ig \implies p_2 f = p_2 g$

Since (p_1, p_2) is a jointly monic pair, we have that $f = g$. Thus, i is a mono. Since i and h are both monos, we have that ih is a mono.

By the following commutative diagram

$$\begin{array}{ccccccc}
 & & & \xi & & & \\
 & & & \curvearrowright & & & \\
 & & E \times_B C & \xrightarrow{h} & C \times_B C & \xrightarrow{i} & C \times C \\
 & & & & p_1 \nearrow & \uparrow p'_1 & \\
 & & & & & C & \\
 & & & & p_2 \searrow & \downarrow p'_2 & \\
 & & & & & C & \\
 & & & \curvearrowleft & & & \\
 & & & \text{proj}_{2,C} & & &
 \end{array}$$

we conclude that $ih = \langle \text{proj}_{2,C}, \xi \rangle$, so $\langle \text{proj}_{2,C}, \xi \rangle$ is a mono.

We proved that $\langle \text{proj}_{2,C}, \xi \rangle$ is an internal relation, now we need to show that it is an equivalence, ie it is reflexive, symmetric and transitive.

- Claim: $\langle \text{proj}_{2,C}, \xi \rangle$ is reflexive.

Called η the unit of the monad, we have that $\langle \text{proj}_{2,C}, \xi \rangle \eta_{(C,\gamma)} = \langle 1_C, 1_C \rangle$. In fact $\xi \eta_{(C,\gamma)} = 1_C$ and by the definition of the unit of the monad, we have that $\text{proj}_{2,C} \eta_{(C,\gamma)} = \text{proj}_{2,C} \langle \gamma, 1_C \rangle = 1_C$.

$$\begin{array}{ccccc}
 & C & & & \\
 & \swarrow 1_C & \downarrow \eta_{(C,\gamma)} & \searrow 1_C & \\
 C & \xleftarrow{\text{proj}_{2,C}} & E \times_B C & \xrightarrow{\xi} & C
 \end{array}$$

- Claim: $(\text{proj}_{2,C}, \xi)$ is symmetric.

Consider the following diagram, where $\text{proj}_{1,E \times_B C}$ and $\text{proj}_{2,E \times_B C}$ are the projections of $E \times_B (E \times_B C)$. Let l be the morphism induced by the universal property of the pullback from $1_{E \times_B C}$ and $\gamma \text{proj}_{2,C}$, in particular it is the unique such that

$$\gamma \text{proj}_{2,C} = \text{proj}_{1,E \times_B C} l = \text{proj}_{1,C} (1_E \times \text{proj}_{2,C}) l$$

and

$$\begin{array}{ccccc}
 & & 1_{E \times_B C} = \text{proj}_{2,E \times_B C} l & & \\
 & & \text{---} 1_{E \times_B C} \text{---} & & \\
 E \times_B C & \xrightarrow{\quad l \quad} & E \times_B (E \times_B C) & \xrightarrow{\text{proj}_{2,E \times_B C}} & E \times_B C \\
 \downarrow \text{proj}_{2,C} & & \downarrow 1_E \times_B \text{proj}_{2,C} & & \downarrow \text{proj}_{2,C} \\
 & & E \times_B C & \xrightarrow{\text{proj}_{2,C}} & C \\
 & \nearrow \eta_{(C,\gamma)} & \downarrow \text{proj}_{1,C} & \nearrow \text{proj}_{1,E \times_B C} & \downarrow p\gamma \\
 C & & E & \xrightarrow{p} & B
 \end{array}$$

Define $\sigma = (1_E \times_B \xi)l$. Observe that

$$\text{proj}_{2,C} \sigma = \text{proj}_{2,C} (1_E \times_B \xi) l = \xi \text{proj}_{2,E \times_B C} l = \xi 1_{E \times_B C} = \xi$$

It remains to show that $\xi \sigma = \text{proj}_{2,C}$. First, by the following compositions and the fact that $(\text{proj}_{1,C}, \text{proj}_{2,C})$ is a jointly monic pair

- $\text{proj}_{1,C} (1_E \times_B \text{proj}_{2,C}) l = \text{proj}_{1,E \times_B C} l = \gamma \text{proj}_{2,C}$
- $\text{proj}_{1,C} \eta_{(C,\gamma)} \text{proj}_{2,C} = \text{proj}_{1,C} \langle \gamma, 1_C \rangle \text{proj}_{2,C} = \gamma \text{proj}_{2,C}$
- $\text{proj}_{2,C} (1_E \times_B \text{proj}_{2,C}) l = \text{proj}_{2,C} \text{proj}_{2,E \times_B C} l = \text{proj}_{2,C} 1_{E \times_B C} = \text{proj}_{2,C}$
- $\text{proj}_{2,C} \eta_{(C,\gamma)} \text{proj}_{2,C} = \text{proj}_{2,C} \langle \gamma, 1_C \rangle \text{proj}_{2,C} = 1_C \text{proj}_{2,C} = \text{proj}_{2,C}$

we have that $\eta_{(C,\gamma)} \text{proj}_{2,C} = (1_E \times_B \text{proj}_{2,C}) l$. Hence, since by *Lemma 1* ξ co-equalizes $(1_E \times_B \xi, 1_E \times_B \text{proj}_{2,C})$, it holds that

$$\xi \sigma = \xi (1_E \times_B \xi) l = \xi (1_E \times_B \text{proj}_{2,C}) l = \xi \eta_{(C,\gamma)} \text{proj}_{2,C} = 1_C \text{proj}_{2,C} = \text{proj}_{2,C}$$

as claimed. Thus the following diagram commutes.

$$\begin{array}{ccccc}
 & & E \times_B C & & \\
 & \swarrow \xi & \downarrow \sigma = (1_E \times_B \xi)l & \searrow \text{proj}_{2,C} & \\
 C & & & & C \\
 & \nwarrow \text{proj}_{2,C} & & \nearrow \xi & \\
 & & E \times_B C & &
 \end{array}$$

- Claim: $(\text{proj}_{2,C}, \xi)$ is transitive.

By (CITE) the pullback of ξ and $\text{proj}_{2,C}$ is the square

$$\begin{array}{ccc}
 E \times_B (E \times_B C) & \xrightarrow{1_E \times_B \xi} & E \times_B C \\
 \downarrow \text{proj}_{2, E \times_B C} & & \downarrow \text{proj}_{2,C} \\
 E \times_B C & \xrightarrow{\xi} & C
 \end{array}$$

We aim to prove that there exists a morphism $\tau : E \times_B (E \times_B C) \rightarrow E \times_B C$ such that

- $\xi(1_E \times_B \xi) = \xi\tau$
- $\text{proj}_{2,C}\text{proj}_{2, E \times_B C} = \text{proj}_{2,C}\tau$

Define $\tau = (1_E \times_B \text{proj}_{2,C})$, we have that

- $\xi(1_E \times_B \xi) = \xi(1_E \times_B \text{proj}_{2,C})$
- $\text{proj}_{2,C}\text{proj}_{2, E \times_B C} = \text{proj}_{2,C}(1_E \times_B \text{proj}_{2,C})$

In particular the first condition holds by *Lemma 1*, and the second one holds by the definition of $(1_E \times_B \text{proj}_{2,C})$

□

In an exact category each equivalence relation is a kernel pair and by (CITE) has a coequalizer. Thus in $\mathcal{A}$ exact category $(\text{proj}_{2,C}, \xi)$, by *Lemma 2* has a coequalizer. Let $q : C \rightarrow Q$ be such coequalizer. Observe that, considering $\text{proj}_{2,C}$ and ξ as morphism in $(\mathcal{A} \downarrow B)$, then q is a morphism in $(\mathcal{A} \downarrow B)$ as well by the universal property. So, called δ the morphism induced by the universal property of the coequalizer from $\text{proj}_{2,C}$ and ξ , we have that q with codomain (Q, δ) is also the coequalizer of the pair in $(\mathcal{A} \downarrow B)$.

$$\begin{array}{ccccc}
 E \times_B C & \xrightarrow{\xi} & C & \xrightarrow{q} & Q \\
 & \searrow \text{proj}_{2,C} & \downarrow p\gamma & \swarrow \delta & \\
 & & B & &
 \end{array}$$

Consider the comparison functor $K : (\mathcal{A} \downarrow B) \rightarrow (\mathcal{A} \downarrow E)^T$ defined as $K(D, \zeta) = (p^*(D, \zeta), p^*(\epsilon_{(D, \zeta)}))$, where ϵ is the counit of the adjunction, and $\epsilon_{(D, \zeta)}$ is defined as the projection from $E \times_B D$ to D of the pullback of p along ζ .

In our context of exact category, turns out that the comparison functor K has a left adjoint $L : (\mathcal{A} \downarrow E)^T \rightarrow (\mathcal{A} \downarrow B)$ defined as $L(C, \gamma; \xi) = (Q, \delta)$, where (Q, δ) is the codomain of coequalizer of the pair $(\text{proj}_{2,C}, \xi)$ in $(\mathcal{A} \downarrow B)$.

$$\begin{array}{ccc}
 (\mathcal{A} \downarrow B) & \xrightleftharpoons[L]{K} & (\mathcal{A} \downarrow E)^T \\
 \uparrow p! & \nearrow F^T & \\
 \downarrow p^* & \nwarrow U^T & \\
 (\mathcal{A} \downarrow E) & &
 \end{array}$$

Let us introduce the counit ρ and counit α of the adjunction $K \dashv L$. For each object (D, ζ) in $(\mathcal{A} \downarrow B)$ the counit is a morphism of the kind $\rho_{(D, \zeta)} : LK(D, \zeta) \rightarrow (D, \zeta)$. First we compute $K(D, \zeta)$.

$$K(D, \zeta) = (E \times_B D, \text{proj}_{1,D}; p^*(\text{proj}_{2,D}))$$

$$\begin{array}{ccc}
 E \times_B D & \xrightarrow{\text{proj}_{2,D}} & D \\
 \downarrow \text{proj}_{1,D} & & \downarrow \zeta \\
 E & \xrightarrow{p} & B
 \end{array}$$

Then, call $q' : E \times_B D \rightarrow Q'$ the coequalizer of the equivalence relation $(\text{proj}_{2,E \times_B D}, p^*(\text{proj}_{2,D}))$, as done above we make it a coequalizer in $(\mathcal{A} \downarrow B)$ via the universal property. We obtain $q' : (E \times_B D, p\text{proj}_{1,D}) \rightarrow (Q', \delta')$. Then compute $LK(D, \zeta)$. By definition of the pullback functor, the top left square in the diagram below commutes. Thus we can define, as shown in the diagram, the morphism $\rho_{(D, \zeta)}$ by the universal property of the coequalizer q' .

$$\begin{array}{ccccc}
 E \times_B (E \times_B D) & \xrightleftharpoons[p_{\text{proj}_{2,E \times_B D}}]{p^*(\text{proj}_{2,D})} & E \times_B D & \xrightarrow{q'} & Q' \\
 \downarrow \text{proj}_{2,E \times_B D} & & \downarrow \text{proj}_{2,D} & & \downarrow \delta' \\
 E \times_B D & \xrightarrow{\text{proj}_{2,D}} & D & \xleftarrow{\rho_{(D, \zeta)}} & \\
 \downarrow \text{proj}_{1,D} & \nearrow \text{proj}_{1,D} & \downarrow \zeta & & \\
 E & \xrightarrow{p} & B & &
 \end{array} \tag{1}$$

Observe that δ' was also defined as the morphism induced by the universal property of the coequalizer, considering the object $E \times_B D$ as an object in $(\mathcal{A} \downarrow B)$ via $p\text{proj}_{1,D}$. Thus $\rho_{(D, \zeta)}$ is a morphism in $(\mathcal{A} \downarrow B)$, ie $\zeta \rho_{(D, \zeta)} = \delta'$.

For each object $(C, \gamma; \xi)$ in $(\mathcal{A} \downarrow E)^T$ the unit is a morphism of the kind $\alpha_{(C, \gamma; \xi)} : (C, \gamma; \xi) \rightarrow KL(C, \gamma; \xi)$. We compute $KL(C, \gamma; \xi)$. Let $q : C \rightarrow Q$ be the coequalizer of the equivalence relation $(\text{proj}_{2,C}, \xi)$ in $\mathcal{A}$. By the universal property of the coequalizer, we have that q is a morphism in $(\mathcal{A} \downarrow B)$ via $p\gamma$. Thus, we can consider the coequalizer of the equivalence relation $(\text{proj}_{2,C}, \xi)$ in $(\mathcal{A} \downarrow B)$, which we denote by $q : (C, p\gamma) \rightarrow (Q, \delta)$. By functoriality, we have that $p^*(q)p^*(\xi) = p^*(q)p^*(\text{proj}_{2,C})$. Moreover, it holds that ξ is the coequalizer of $(p^*(\xi), p^*(\text{proj}_{2,C}))$ by *Lemma 1*. Thus we can define $\alpha_{(C, \gamma; \xi)}$ as the morphism induced by the universal property of the coequalizer q , as done in the diagram below. One can check that $\alpha_{(C, \gamma; \xi)}$ is a morphism in $(\mathcal{A} \downarrow E)^T$.

$$\begin{array}{ccccc}
E \times_B (E \times_B C) & \xrightarrow[p^*(\text{proj}_{2,C})]{p^*(\xi)} & E \times_B C & \xrightarrow{p^*(q)} & E \times_B Q \\
\downarrow \text{proj}_{2, E \times_B C} & & \downarrow \text{proj}_{2,C} & \searrow \alpha_{(C, \gamma; \xi)} & \downarrow \text{proj}_{2,Q} \\
E \times_B C & \xrightarrow[\text{proj}_{2,C}]{\xi} & C & \xrightarrow{q} & Q \\
\downarrow \text{proj}_{1,C} & \nearrow \gamma & \downarrow p\gamma & \nearrow \delta & \\
E & \xrightarrow{p} & B & &
\end{array} \tag{2}$$

One can show that ρ and α are natural transformations, and that they satisfy the universal property of the counit / unit. This follows from the universal property of the coequalizer of the equivalence relation considered.

Our goal is to show that p^* is a monadic functor, ie that the comparison functor K is an equivalence of categories. We prove that it is an adjoint equivalence, which is the same. In order to do so, we aim to prove that the unit α and the counit ρ of the adjunction $L \dashv K$ are pointwise isomorphisms.

Proposition 1. *In an exact category if $p : E \rightarrow B$ is a regular epi, then the counit ρ of the adjunction $L \dashv K$ is a pointwise isomorphism.*

Proof. We aim to prove that for each object (D, ζ) in $(\mathcal{A} \downarrow B)$, the morphism $\rho_{(D, \zeta)} : LK(D, \zeta) \rightarrow (D, \zeta)$ is an isomorphism. Consider the diagram (1) of the definition of the counit $\rho_{(D, \zeta)}$. An exact category is also regular, in particular regular epi are pullback stable. If p is a regular epi, then $\text{proj}_{2,D}$ is a regular epi as well. The top left square is a pullback by (CITE), thus $(p^*(\text{proj}_{2,D}), \text{proj}_{2, E \times_B D})$ is the kernel pair of $\text{proj}_{2,D}$. Since $\text{proj}_{2,D}$ is a regular epi, then, by (CITE), it is the coequalizer of its kernel pair $(p^*(\text{proj}_{2,D}), \text{proj}_{2, E \times_B D})$. We have that q' and $\text{proj}_{2,D}$ are coequalizers of the same pair, thus the morphism $\rho_{(D, \zeta)}$ defined by the universal property of the coequalizer q is actually an isomorphism. $\square$

Proposition 2. *In an exact category the unit α of the adjunction $L \dashv K$ is a pointwise isomorphism.*

Proof. We aim to prove that for each object (C, γ, ξ) in $(\mathcal{A} \downarrow B)$, the morphism $\rho_{(D, \zeta)} : LK(D, \zeta) \rightarrow (D, \zeta)$ is an isomorphism. Consider the diagram (2). An exact category is also regular, in particular regular epi are pullback stable. Since q is a regular epi, then $p^*(q)$ is a regular epi as well. Moreover, since q is the coequalizer of $(\xi, \text{proj}_{2,C})$ which is a kernel pair, then by (CITE) $(\xi, \text{proj}_{2,C})$ is the kernel pair of q . Then, since p^* is a right adjoint, it preserves kernel pairs, thus $(p^*(\xi), p^*(\text{proj}_{2,C}))$ is the kernel pair of $p^*(q)$. We already proved that $p^*(q)$ is a regular epi, then by (CITE) it is the coequalizer of its kernel pair $(p^*(\xi), p^*(\text{proj}_{2,C}))$. By *Lemma 1* also ξ is the coequalizer of $(p^*(\xi), p^*(\text{proj}_{2,C}))$. We have that $p^*(q)$ and ξ are coequalizers of the same pair, thus the morphism $\alpha_{(C, \gamma; \xi)}$ defined by the universal property of the coequalizer ξ is actually an isomorphism. $\square$

Finally, we have the main result we were aiming to prove.

Theorem 1. *In an exact category, if $p : E \rightarrow B$ is a regular epi, then p^* is a monadic functor.*

Proof. The exactness of the category implies that the comparison functor $K : (\mathcal{A} \downarrow B) \rightarrow (\mathcal{A} \downarrow E)^T$ has a left adjoint $L : (\mathcal{A} \downarrow E)^T \rightarrow (\mathcal{A} \downarrow B)$, as shown. Since p is a regular epi, the unit α and the counit ρ of the adjunction $L \dashv K$ are pointwise isomorphisms, by *Proposition 1* and *Proposition 2*. Thus, $L \dashv K$ is an adjoint equivalence. This implies that K is an equivalence of categories. So $p^* : (\mathcal{A} \downarrow B) \rightarrow (\mathcal{A} \downarrow E)$ is a monadic functor. $\square$

References

- [JT91] G. Janelidze and W. Tholen. “How algebraic is the change-of-base functor?” In: *Category Theory, proceedings of the International Conference held in Como, Italy, July 22-28, 1990*. Lecture Notes in Mathematics 1488. Berlin and Heidelberg: Springer, 1991, pp. 174–186.
- [ST92] M. Sobral and W. Tholen. “Effective descent morphisms and effective equivalence relations”. In: *Category Theory 1991, proceedings of an International Summer Category Theory Meeting held June 23-30, 1991*. Canadian Mathematical Society Conference Proceedings 13. Providence, RI: American Mathematical Society, 1992, pp. 421–433.